

T&M equipment for WiMAX

1. EDA software: Agilent's ADS
2. Simulation software: AWR's Visual System Simulator (VSS)
3. Vector signal generators: Agilent's E4438C (ESG), N5182A (MXG) and PSG, R&S SMU200A
4. Signal analysers and software tools: R&S FSQ, Agilent's N5106A, Tektronix RSA 6000, RSA-IQWiMAX, N7613A and N7615A Signal Studio, Tektronix G35
5. Conformance testers: R&S TS8970, N6434A, E6651A, N6430A
6. One-box solutions: R&S CMW270, Agilent's N8300A

"The forum precisely defines mandatory and optional test cases that every device has to face," shares Basavaraju. Conformance tests are done to verify the interoperability of devices and network at various interfaces.

On the device side. WiMAX Forum has associated with AT4 Wireless, a designated certification laboratory and the WiMAX Forum lead lab. All the required testing services for obtaining WiMAX certification for both the base stations and mobile stations are currently available at AT4 Wireless' WiMAX Laboratory.

AT4 Wireless has developed MiNT (mobile communications integrated tester)—a family of test platforms for mobile communications. MiNT T2110 platform covers the official test cases for certification of base stations and subscriber stations according to the WiMAX Mobile radio test specification and supports the initial profiles by the WiMAX Forum.

The network conformance test (NCT) system provides a validated test regime for 'testing and test control notation version-3' (TTCN-3) network conformance testing. Network conformance and protocol conformance test (PCT) systems can be combined together as these use the same test infrastructure—WiMAX certification test program.

The radiated performance test (RPT) is undergoing evaluations and will be a part of certification testing for Mobile WiMAX. At the RPT testing facility installed by AT4 Wireless, one can perform both 2-D (polar) and 3-D (spherical) antenna pattern measurements of transmit power and receiver sensitivity for notebooks, PC-MCIA cards and other terminals with

WiMAX. This will help to ensure the quality of the product.

On the software side. WiMAX systems are built around an IP core network consisting of access service network and connectivity service network. It runs on WiMAX protocols, which are tested using a protocol analyser. "The aim of the WiMAX network is to accommodate the maximum number of users and every user expects the best quality of service," says Sarpotdar. The final end-user experience depends on all the elements of the network performing to their specifications.

For protocol conformance test, TTCN-3 provides the scripting language. The tester should provide protocol designers with a TTCN-3 scripting interface that supports script development for design needs, as well as the ability to run the European Telecommunications Standards Institute (ETSI) validated protocol test cases in accordance with the WiMAX Forum PCT certification requirements.

Talking about the security aspect, Poliecek says, "Another critical area where testing is needed is security and its implementation within the IEEE 802.16e framework. The need for security is increasing, as is subscribers' demand for it. Security solutions emerging among new cellular standards are increasingly complex." He adds, "WiMAX technology needs to interwork flawlessly with other technologies including 3G and its evolutions. This adds yet another dimension to the development test puzzle."

One-box test systems

Many T&M manufacturers believe in providing an integrated test solution as it is easier to use for engineers and design-

ers. Typically, a one-box solution for the WiMAX system has a signal waveform generator, a signal analyser and some software tools that provide measurement and analysis of the system.

Agilent's N8300A tester combines the capabilities of both a signal generator and a signal analyser in one box. It incorporates software customisations to test and calibrate specific WiMAX chipsets from different manufacturers. The N8300A is optimised for testing WiMAX radios during manufacturing.

CMW270, a complete test product from Rohde & Schwarz, was chosen as the 'Best in Test' in 2008 by the readers of *Test and Measurement World*, a trade publication and website. It is an all-in-one solution for testing mobile WiMAX equipment accurately and fast. It allows users to select from a non-signaling to the signaling mode to test a base station emulator in simulated network condition.

User experience matters

"The end-user experience is important. Thus it is important to perform quality-of-service tests to check for throughput, packet errors and latency. A complete network check is necessary to ensure that there are no dark spots and that all the areas have adequate signal strength," says Sarpotdar.

The network frequently needs to grant high data rates to subscribers attempting to run throughput-intensive applications. Developers must have a means to guarantee data capture and data processing performance up to the maximum WiMAX base station throughput.

Every manufacturer has new software and instrumentation that increase the bandwidth to process wider-bandwidth signals for WiMAX. New T&M equipment and software dedicated to RF and WiMAX testing greatly reduce the development and deployment time. These reduce the manufacturing and maintenance costs as well and help improve the user experience. ●

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